Algorithms and Data Structures 1

Project 2026-2027

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Introduction

The goal is to write a program that manages the customers of a hotel chain.

Assume that the chain has several locations in different cities, and that each location accommodates a number of customers.

All data and the links between these data items will be represented using singly linked lists.

A. Structures

1. Create a **Client** structure (and the corresponding type **T_Client**) with the following fields:

   last_name: a string
   first_name: a string
   age: an integer
   num_room: an integer
   next: a pointer to a Client structure

2. Create a **Site** structure (and the corresponding type **T_Site**) with the following fields:

   city: a string
   number_clients: an integer (the number of current bookings at this location)
   first: a pointer to a Client structure
   next: a pointer to a Site structure

3. Create a **Hotel** structure (and the corresponding type **T_Hotel**) with the following fields:

   name: a string
   first: a pointer to a Site structure

B. Required Functions

1. Creating and initializing structures:

```
T_Client *createClient(char *last_name, char *first_name, int age, int num_room)
T_Site *createSite(char *city)
T_Hotel *createHotel(char *name)
```

2. Adding a booking at a location:

```
int addClient(T_Site *site, T_Client *client)
```

Return 1 if the insertion succeeds, or 0 otherwise. Insert the client while preserving alphabetical order by the client's last name and then first name.

3. Adding a location to the hotel chain:

```
int addSite(T_Hotel *hotel, T_Site *site)
```

Return 1 if the insertion succeeds, or 0 otherwise. Insert the location while preserving alphabetical order by city name. The user must not be allowed to add the same city twice.

4. Displaying all locations:

```
void displayHotel(T_Hotel *hotel)
```

Display the locations as a list sorted by city name.

City	Number of customers
Bordeaux	63
Lille	58
Paris	123

5. Displaying all customers at a location:

```
void displaySite(T_Site *site)
```

Display the customers in a table sorted by last name and first name.

Last name	First name	Age	Room
Dupond	Charles	28	125
Dupond	Michele	46	56
Martin	Alain	42	112

6. Deleting a booking at a location:

```
int deleteClient(T_Site *site, char *last_name, char *first_name)
```

This function returns 1 if deletion succeeds, or 0 otherwise. Make sure to free the memory that was previously allocated.

7. Deleting a location and all customers it contains:

```
int deleteSite(T_Hotel *hotel, char *city)
```

The function returns 1 if deletion succeeds, or 0 otherwise. Make sure to free the memory that was previously allocated.

8. Searching for customers whose last name starts with the string supplied as a parameter:

```
void searchClients(T_Hotel *hotel, char *critereaName)
```

Display the results in a table sorted by ***last name*** and ***first name***.

Last name	First name	Age	Room	City
DUPOND	Agathe	23	12	Bordeaux
DUPOND	François	35	29	Paris
DUPONT	Amélie	28	17	Lille
DURAND	Antoine	42	15	Paris
DUVAL	Magali	40	38	Bordeaux

In your report, explain whether you chose to create specific structures or adapt the structures already defined to implement this function.

9. Processing a booking file

```
void processingfile(T_Hotel *hotel)
```

Your program must read a text file in the format shown below and add one booking for each line, creating the relevant location if necessary.

```
LastName;FirstName;Age;Room Number;City
ANTOINE;Michel;40;38;Bordeaux
MARTIN;Lucie;32;45;Lille
DUPOND;Antoine;42;25;Toulouse
LECLERC;Denis;25;112;Lyon
...
```

C. Main Program

Implement a menu containing the following options:

- 1. Create a hotel chain
- 2. Add a location to a hotel chain
- 3. Add a booking at a location
- 4. Display the locations in a hotel chain
- 5. Display the customers at a location
- 6. Delete a booking at a location
- 7. Delete a location
- 8. Search for a customer by last name
- 9. Quit

General Instructions

Source files

Organize your project as follows:

- Header file tp3.h, containing type definitions, structures, constants, and function prototypes.
- Source file tp3.c, containing the function definitions.
- Source file main.c, containing the main program.

Report

Your report must be no more than four pages and must contain:

- A list of the additional structures and functions you chose to implement, with the reasons for those choices.
- A brief explanation of the time complexity of each implemented function.

Your report and the three source files must be submitted as an assignment on Moodle, in the submission area provided for this purpose (only one submission per team). DON'T MISS THE DEADLINE!

A presentation of your program and some Q/A about your project will be done by every team at the last exercise sessions!